Examiner only

6. (a) Enzymes are described as biological catalysts. State the function of enzymes in cells. [1]

....

....

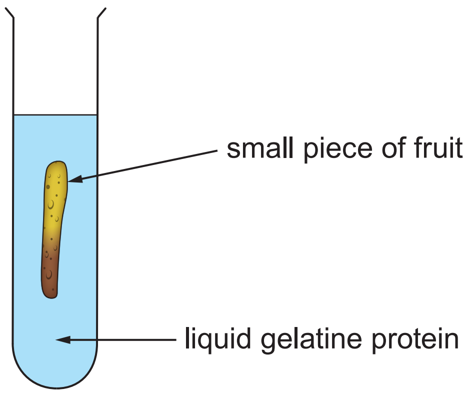
(b) Gelatine is a protein which is liquid above 25 °C. The gelatine protein sets when it cools down and becomes fully solid at 15 °C. The gelatine protein does not set if it is broken down by an enzyme. **Image 6.1** shows a packet of gelatine.

Image 6.1

Many fruits have protease enzymes which break down proteins.

In an investigation some students used the gelatine protein to identify which fruits contain proteases.
They set up five test tubes, one of which is shown in **Image 6.2**.

Image 6.2



Examiner only

They placed all the tubes in a refrigerator at 5 °C for some time. They then observed the tubes. Their results are shown in **Table 6.3**.

Table 6.3

		Gelatine protein (liquid or solid)	
Tube number	Fruit	At start	At end
1	fresh figs	liquid	liquid
2	fresh strawberry	liquid	solid
3	fresh kiwi fruit	liquid	liquid
4	boiled peaches	liquid	solid
5	fresh pineapple	liquid	liquid

(i) Use the information about gelatine and **Table 6.3** to answer the following questions.

I. Identify **all** the fruits which contain protease. [1]

....

II. Explain why you reached this conclusion. [2]

....

....

(ii) The students' teacher commented that the result for peaches was not valid.

I. Explain the reason for this comment. [2]

....

....

....

....

II. State how they could obtain a valid result for peaches. [1]

....

....

(iii) Suggest a temperature at which gelatine should be kept before pouring it into the test tubes. [1]

.... °C

(iv) State **one** variable, other than temperature, which the students should have controlled to ensure fair testing. [1]

....

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